Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

4bit

Selected Domain

Domain 1

Selected SDG Track (1–17)

SDG 4.3, 4.4, 8.2, & 8.5

Project / Solution Name

Skilytics

Name (Leader)

Muhammad Ameer Yusoff Bin Azeri

Email & Phone (Leader)

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University Affiliation(s)

Universiti Teknologi Malaysia

Number of Team Members

4

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

As university students, we witness on our own how industry skill demands change in months due to the rapid growth of AI, though academic and TVET syllabuses take years to adapt. Access to high-quality and market-aligned higher education is severely limited by this "curriculum lag" (SDG 4.3). As a result, graduates and job seekers enter the workforce with outdated qualifications rather than the relevant technical and professional skills required by current companies (SDG 4.4). The entire job ecosystem is affected by the crisis. Without flexible reskilling pathways, working individuals are at risk of displacement, and career centers lack real-time labor market data to successfully mentor students. A negative contradiction is created by this disconnect, causing significant talent shortages in high-growth industries to coexist with rising graduate unemployment and job fragility. The gap negatively impacts full and productive employment by failing to train workers for respectable and durable jobs (SDG 8.5). Additionally, ongoing skill gaps hinder workforce productivity and technological innovation, which directly lowers national economic productivity (SDG 8.2). Learners are forced to choose between static curricula and an AI-driven economy in the absence of a real-time bridge connecting university education and vocational training to live industry expectations while endangering both individual livelihoods and overall economic growth.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Skilytics is an AI-powered career guidance and reskilling platform that helps workers stay relevant by aligning their current skills with planned upskilling and reskilling to meet real job market demands. The platform uses a step-by-step AI-powered workflow that automatically extracts skills from uploaded CVs, runs a Holland Code (RIASEC) interest test, and compares user profiles against standardized O*NET job requirements. Specialized AI agents then evaluate how exposed a user's current role is to AI knowledge, identify specific skill gaps in the local market, and generate a personalised learning plan matched to trusted courses.

Through the SDC Phase 1 workshop, we learned how to quickly create and test apps using an online workspace. I gained hands-on experience using Replit to speed up writing code, fix mistakes, and build working web tools in a short time. I also learned how to easily launch these projects online so others can use them right away without needing to install complicated software on my computer.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

- Frontend & UI Layer
  - Core Framework: React with TypeScript, built using Vite.
  - Styling & Components: Tailwind CSS paired with Shadcn UI component libraries.
  - Hosting: Vercel.
- Backend & API Layer
  - Runtime & Framework: Node.js with Express in TypeScript.
  - Validation & Parsing: Zod for resume text extraction.
  - Hosting: Render.
- Database & ORM
  - Database: Neon Serverless PostgreSQL.
  - Data Access Layer: Drizzle ORM for schema management, migrations, and database queries.
  - Stored Data: User profiles, psychometric assessment results, resume data, and custom reskilling plans.
- Artificial Intelligence & Machine Learning
  - Cloud LLMs: Gemini-flash-lite 3.1
  - Open-Source Models: Qwen 3.5-27b, & Qwen3.5-9b hosted on RunPod Serverless GPU Endpoints
- External Data Integrations
  - Malaysian Labour Statistics (OpenDOSM): Fetches monthly labour force survey data from data.gov.my to power local market and graduate outlooks.
  - AI Economic & Task Exposure Data (Anthropic Economic Index via Hugging Face): Pulls two datasets about country-level AI usage and occupation-level task exposure by SOC code.
  - Occupation & Skill Taxonomy (O*NET Database v31.0): Integrates 30 official US occupation, skill, and interest profiles mapped via a Standard Occupational Classification (SOC) crosswalk.
  - Training & Course Providers (External Link Verification): Tracks and runs automated availability checks against external provider URLs.

Yes, our project used official SDC sponsor platforms, specifically Replit and RunPod. We used Replit to build the foundation of our web app, including creating the user interface, developing the core features, and connecting the database. We also used RunPod's serverless GPUs to run our AI models quickly and efficiently on demand.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our three target scopes have very different needs, so we built tailored interfaces for each:

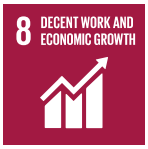
- Enterprises: A detailed, professional dashboard focused on current and future workforce planning, tracking employee AI exposure, and finding HRD-claimable training.
- Educators: A comprehensive, data-rich portal to track present and future student development, monitor cohort AI exposure, and align courses with industry trends.
- Individuals: A simple, ELI5 (Explain Like I'm 5) interface that cuts out clutter to show only clear upskilling and reskilling steps alongside personal present and future AI exposure tracking.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	 Target 4.4 "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employmAent, decent jobs and entrepreneurship." Target 4.3 "By 2030, ensure equal access for all women and men to affordable and quality technical, vocational and tertiary education, including university."  Target 8.2 "Achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including through a focus on high-value added and labour-intensive sectors." Target 8.5 "By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	In Malaysia, 37.3% of tertiary-educated workers are currently affected by skill-related underemployment, while up to 44% of core workplace skills are projected to be disrupted by AI and automation within the next five years.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Assessment efficiency: % reduction in time taken to assess candidate skill gaps and generate a personalized reskilling roadmap. Processing Speed: % reduction in end-to-end processing time for the multi-agent AI pipeline Recommendation Accuracy: % accuracy in matching extracted candidate skills to standardized job roles and relevant courses.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	80% faster career profiling: Reduce the end-to-end time needed to parse CVs, assess skills against market roles, and generate a certified learning path from days of manual guidance to under 2 minutes. 60% reduction in AI processing time, cutting the full pipeline evaluation from over 3 minutes down to under 45 seconds per profile. 85% accuracy in role and course recommendations, verified against benchmark job profiles and standard career datasets.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
OpenDOSM Monthly Labour Force	Dataset	CC BY 4.0
AI Usage by Country (Anthropic Economic Index)	Dataset	CC BY 4.0
AI Exposure by Occupation (Anthropic Economic Index)	Dataset	CC BY 4.0
O*NET Occupation References	Dataset	CC BY 4.0
Microsoft Learn Public Course Catalogue	Dataset	Microsoft Terms of Use

Name	Type (code/dataset/model)	License
freeCodeCamp Public Curriculum	Dataset	CC BY-SA 4.0 / BSD 3-Clause
GitHub Skills Exercises	Dataset	MIT / GitHub Terms
Hugging Face Learn	Dataset	Open Access / CC BY 4.0
Holland Code (RIASEC) Question Set	Dataset	Public Domain
HubSpot Academy Public Catalogue	Dataset	HubSpot Terms of Service
DOSM JMI Q4 2025 · Selected Occupation Snapshot	Dataset	CC BY 4.0 / Open Data
Google News Public Career RSS	Dataset	Google Terms of Service
Gemini-flash-lite 3.1	Model	Google APIs Terms of Service
Qwen 3.5-27b Qwen3.5-9b	Model	Apache 2.0 / Qwen Open License

Name	Type (code/dataset/model)	License
BLS US Career Growth Reference (2025–2035)	Dataset	Public Domain (U.S. Govt)
O*NET Content Model · Skill References	Dataset	CC BY 4.0
United Nations · SDG Framework	Dataset	UN Open License / CC BY 3.0 IGO
OpenLearn · Reviewed Public Courses	Dataset	CC BY-NC-SA 4.0 / Open University Terms
Himalayas · Public Vacancies	Dataset	Himalayas Terms of Use / Keyless API
Remotive · Public Remote Vacancies	Dataset	Remotive API Terms / Keyless API
RemoteOK · Public Job Feed	Dataset	RemoteOK Public Terms / Keyless API

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

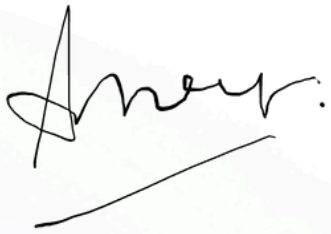
☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

24/10/2026